

AI Incidents and Ethics

Racial Bias in a Healthcare Risk Algorithm

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Case Study Selection and Summary

In October 2019, researchers reported racial bias in a widely used healthcare risk algorithm that helped health systems identify patients for extra care. The algorithm did not use race directly; instead, it predicted future healthcare spending as a proxy for health need.

Because less money was historically spent on Black patients with similar levels of illness, the system underestimated their needs. Researchers found that correcting this proxy would more than double the share of Black patients identified for additional care. The case showed how apparently neutral data can reproduce existing inequities in high-stakes decisions.

Critical Analysis: Three Ethical Issues

ISSUE 1

Unequal Access to Additional Care

Why I chose it. This was the clearest harm in the case. Patients with similar medical needs could be treated differently because the model converted an unequal healthcare system into a prediction target.

My thought process. At first I expected bias to come from a sensitive field such as race being included in the data. What changed my thinking was that race was excluded, yet bias still appeared because healthcare cost was used as a stand-in for health need. That proxy reflected differences in access and spending rather than differences in how sick patients were. It showed me that removing a protected attribute does not make a model fair if the target itself carries historical inequality.

Proposed solution. Define the prediction target around actual health need, such as clinical conditions and outcomes, rather than future spending alone. Before deployment, developers should also compare error rates and eligibility outcomes across demographic groups to verify the model is not systematically withholding care.

ISSUE 2

A Misleading Proxy for Health Need

Why I chose it. The model was not simply making a bad prediction; it was optimizing the wrong measure for the decision being made. Cost was convenient and measurable, but it was not ethically equivalent to medical need.

My thought process. This made me think more carefully about how a technically reasonable target can become an ethical problem. A model can perform well against the label it was trained to predict and still produce harmful decisions if that label does not represent the real goal. Predicting cost accurately could still understate the needs of patients who historically received less care. The failure therefore started before model training, when the problem and target variable were defined.

Proposed solution. Require a documented justification for every high-impact proxy variable and test whether it behaves differently across affected groups. Model validation should include domain experts who can challenge whether the chosen target actually represents the human outcome the system is supposed to improve.

Transparency, Oversight, and Accountability

Why I chose it. The algorithm influenced who received additional healthcare resources, yet patients could not realistically evaluate how that prioritization was produced. High-stakes decision support requires stronger oversight than an ordinary business prediction.

My thought process. Human involvement alone does not guarantee meaningful oversight. Clinicians may trust a risk score because it appears objective, especially when the model is already embedded in a workflow. If the model's target, limitations, and subgroup performance are not visible, users may not know when to question its recommendation. Responsibility can then become blurred between the vendor, the health system, and the clinician.

Proposed solution. Require independent bias audits, clear documentation of model purpose and limitations, and ongoing monitoring after deployment. The organization using the model should also assign a named owner responsible for reviewing harms and suspending the system when performance becomes unsafe or inequitable.

Reflection on Critical Thinking

When I started this case, I assumed algorithmic bias mainly came from biased training data or from directly including a sensitive attribute such as race. My thinking changed when I saw that this model excluded race and could still create racial disparities because healthcare cost was used as a proxy for health need. I now see that fairness has to be considered when defining the problem, choosing the target, validating the model, and monitoring it after deployment. This will influence how I approach AI and data systems in my own work. I do not want to stop at asking whether a model is accurate or whether a pipeline is technically correct. I will also ask what the target represents, whose experience may be missing from the data, and whether errors are distributed unevenly across groups before I trust the output.

Sources Used to Validate the Case Study

- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453. <https://doi.org/10.1126/science.aax2342>
- National Library of Medicine. (2019). PubMed record for Dissecting racial bias in an algorithm used to manage the health of populations (PMID 31649194). <https://pubmed.ncbi.nlm.nih.gov/31649194/>
- AI Incident Database. (2019). Incident 124: Optum Algorithmic Health Risk Scores Reportedly Underestimated Black Patients' Needs. <https://incidentdatabase.ai/cite/124/>

UC Berkeley Center for Health Management and Innovation. (2019, October 25). Dissecting racial bias in an algorithm used to manage the health of populations. <https://chmi.berkeley.edu/news/dissecting-racial-bias-in-an-algorithm-used-to-manage-the-health-of-populations>

AI disclosure: I used an AI tool to assist with research direction, organization, checking the assignment structure and word limits, and improving grammar and clarity. I reviewed the sources and revised the analysis and reflection to reflect my own understanding.